Adaptive SAR Image Enhancement for Aircraft Detection via Speckle Suppression and Channel Combination

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***Abstract*— Synthetic aperture radar (SAR) possesses significant advantages in aircraft detection due to its all-day and all-weather monitoring capability, but some unique problems in SAR images decrease the performance of aircraft detection. The speckle effect and excessive dynamic range are the most common problems that interfere with the visual features in SAR images and deteriorate detection performance. However, there lacks a detection-oriented image enhancement algorithm to collaboratively solve these two problems. An adaptive image enhancement algorithm is proposed to improve the performance of aircraft detection in SAR images. The proposed image enhancement algorithm provides a pseudocolor image through speckle suppression and channel combination, which consists of the speckle noise suppression channel, strong scattering feature enhancement channel, and weak scattering feature enhancement channel. The speckle noise suppression is achieved by a despeckle network, and the radiational feature enhancement channels are derived from an adaptive quantization method based on the characteristics of amplitude distribution. By optimizing the quality of the input image, the proposed image enhancement algorithm improves the performance of aircraft detection. Experiments based on datasets acquired by GaoFen-3 satellites indicate that the proposed algorithms significantly improve the detection performance of various types of detectors. The source project is available at https://github.com/suoyuxi/ChannelEnhancement.**

***Index Terms*— Aircraft detection, channel combination, despeckling, synthetic aperture radar (SAR).**

I. Introduction

SYNTHETIC aperture radar (SAR) acquires high-resolution images by actively transmitting electromagnetic waves and has the ability to capture targets obscured by clouds, fog, and even sand [1]. Benefiting from its ability of all-day and all-weather global monitoring, SAR is fruitfully applied in aircraft detection. Aircraft detection relies on the essential features contained in the SAR images, such as textural and structural features, which play an important role in distinguishing objects and backgrounds. Due to the special imaging mechanism of SAR images, these features are affected by the unique characteristics of SAR images and are quite different from those of optical images. Speckle effect and excessive dynamic range are very common characteristics in SAR images, which profoundly affect the textural and structural features of images. The speckle noise weakens the edges and textural features of targets, and the excessive dynamic range interferes with the effective feature representation of the target and background. Various types of speckle noise suppression algorithms and image quantization methods have been proposed to solve these problems, so as to optimize the image quality and improve the subsequent applications.

The speckle effect is a common problem that interferes with SAR images in the form of multiplicative noise. It makes the background such as the apron cluttered and disturbs the edges and contours of aircraft docked on the apron. Together with the fluctuation of the radar cross section (RCS) under different observation conditions [2], the speckle noise aggravates the instability and fragmentation of the textural and structural features. The speckle noise, hence, seriously deteriorates the performance of aircraft detection. Traditional despeckle algorithms such as the block-matching and 3-D filtering (BM3D) algorithm [3] and the Lee filter algorithm [4] achieve superior signal-to-noise ratio (SNR), which improves the distinction between the target and the background. However, these algorithms do not pay enough attention to the maintenance and restoration of the structural features [5], so the improvement in aircraft detection is limited.

The dynamic range of the SAR image is far beyond that of the optical image, which can reach 50 dB or even higher [6]. The amplitude distribution of SAR images presents a unimodal and long-tailed shape [7], and the long tail rarely carries effective information. As the mainstream detectors require effective visual features and light storage burden, it is common to convert the encoding format into the 8-bit encoding format. One commonly used conversion method truncates the long tail and linearly maps the amplitude values into an 8-bit gray-scale domain. Because of the different shape and scale parameters,

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the converted gray-scale distributions change with observed scenes. The visual features become unstable and inconsistent after quantization and further interfere with the interpretation. Another familiar quantization method is the logarithmic quantization method. It makes the speckle noise easier to cope with but devilishly compresses the strong scattering region due to its rapidly decreasing mapping derivative. The radiational details of high scattering areas and targets, hence, seriously degrade.

Existing optimization methods solve the above problems separately, and there is a lack of a comprehensive image enhancement method for object detection. To achieve collaborative optimization and improve detection performance, this article proposes an adaptive SAR image enhancement algorithm. The core is to design a channel optimization and combination method to improve the extraction of visual features and adequately express the detailed information. The proposed solution consists of the despeckle enhancement transformer network (DET-net) and the adaptive quantization method combined with SAR amplitude distribution characteristics. Three enhanced data channels can be obtained through the mentioned processing methods, which include the multiplicative noise suppression channel, strong scattering feature enhancement channel, and weak scattering feature enhancement channel. After the channel combination, the enhanced pseudocolor image is applied to improve the aircraft detection performance through the optimization of the detector input.

The multiplicative noise suppression channel is derived from DET-Net, which is essentially a codec network. The encoder discards the multiplicative noise as redundant information and constructs the latent representation of the textural and structural features through the analysis of the context information. The decoder translates the latent representation into the image domain and ultimately acquires the enhanced image while suppressing the speckle noise. In addition, the homomorphic transformation and a self-supervised learning method based on the Bernoulli sampling [8] are applied for the model training. The optimization strategy based on the characteristics of the multiplicative noise effectively assists the despeckle model with convergence.

An adaptive quantization method based on the unimodal distribution is proposed to obtain the strong scattering feature enhancement channel and weak scattering feature enhancement channel. The specific amplitude distribution changes with the observed scenes [9], [10], [11], but they basically present the long-tailed unimodal shape. Based on this property, the converted gray-scale distribution can be mapped into a relatively consistent shape, and the stable gray-scale distribution is conducive to the extraction of visual features. In addition, the details of the strong scattering region and the weak scattering region can be separately emphasized by adjusting the shape of the convert results. Generally speaking, the strong scattering feature enhancement channel maintains more information on targets, while the weak scattering feature enhancement channel maintains more information in the background.

The enhanced data channels are combined into a pseudocolor image to achieve collaborative optimization of the image quality. The image enhancement algorithm provides the detector with more stable, prominent, and abundant features of both targets and background, thereby improving the detection effect. Experiments based on GaoFen-3 SAR images indicate that the adaptive image enhancement algorithm for aircraft detection achieves superior detection performance.

The main innovations are summarized as follows.

1) DET-Net is proposed to eliminate the interference of the multiplicative noise and provide more stable and complete features for aircraft detection. The homomorphic transformation is employed to assist DET-Net with noise elimination, and a self-supervised learning method based on the Bernoulli sampling is applied for the optimization.
2) An adaptive quantization method based on the unimodal distribution is employed to convert the encoding format of amplitude images. It gives consideration to the detailed information of both background and targets and provides consistent and stable gray-scale distribution.
3) The channel combination method is proposed to combine the optimization results of the speckle noise suppression and adaptive quantization. The pseudocolor image obtained by the channel combination weakens the interference of speckle noise and enriches the detail information after the quantization. The channel combination method provides more robust and richer information for the optimization and application of the detector.

This article is organized as follows. Section II introduces the research work that is related to the statistical characterization of the SAR image, despeckle algorithms, and aircraft detection. Section III details the adaptive image enhancement algorithm. Section IV carries out experiments based on GaoFen-3 [12], [13] images to verify the effectiveness of the proposed method. Section V concludes this article.

II. Relative Work

A. Statistical Characterization of the SAR Image

The existing mathematical models that describe the statistical characterization of the SAR image can be divided into two categories: the deterministic models and the statistical distribution model.

The deterministic models [14], [15], [16] construct sufficient precise and sophisticated equations for the whole process of microwave propagation and acquire the specific statistical characterization of a specific scenario. Due to the enormous parameters and strict scenario limitation, the deterministic models are rarely applied for images' interpretation [7].

The statistical distribution models aim to implement statistical and mathematical analysis on the speckle characteristics and amplitude distribution [9]. The Rayleigh distribution [9], [17] is a basic model describing the amplitude distribution. Its shortage of textual analysis limits its application for complicated scenarios. The multiplicative models, such as K distribution [7] and inverse Gaussian distribution [18], combine the texture distribution and the speckle effect to further describe heterogeneous images and are widely applied for SAR image interpretation. The specific mathematical form of the statistical distribution model also changes with the observed scene. For example, the lognormal distribution [19]

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is often used to describe forest areas, while the Fisher distribution [20] is more suitable for analyzing high-resolution images of urban areas. Although the specific forms are different, these distributions basically present a long-tailed unimodal shape, which is a unified property applicable to different scenarios.

B. Speckle Noise Suppression

Despeckle algorithms in SAR images can be roughly divided into two categories [21], [22]. Some researches directly transfer the noise suppression algorithm of optical images to SAR images [23], [24]. However, these algorithms are basically proposed for the additive white Gaussian noise (AWGN), which is quite different from the speckle noise. Among these noise suppression algorithms of optical images, the BM3D algorithm [3] is almost the most powerful and representative one. The BM3D algorithm suppresses high-frequency noise by repeatedly using collaborative filtering. First, it applies collaborative filtering to process the 3-D groups that consist of similar 2-D image patches to obtain a preliminary filtering result. In the second collaborative filtering, the preliminary filtering result is used for the refined group matching, and the collaborative shrinkage is used to obtain the final filtered result.

Another type of algorithm specializes in the suppression of speckle noise. Besides rigorous mathematical analysis, they depend on certain assumptions to establish the crucial statistical models. Two subcategories are further generalized based on different assumptions [21].

1) Algorithms based on the multiplicative model assume that the noise is stationary and ergodic. The assumption is the prerequisite for the estimation of statistical characteristics and the construction of corresponding filters. Lee filtering is the most typical one based on the stationary assumption and minimum mean-square error (MMSE) principle. The edge-preserving optimized speckle filter (EPOS) [25] utilizes the relative standard deviation to limit the areas for filtering and performs better on both speckle suppression and edge preservation.
2) Algorithms based on the product model construct the equation under the maximum a posteriori (MAP) probability rule to solve the unspeckled intensity. They require the probability density function (pdf) of the speckle noise as prior knowledge to solve the MAP equation. The common MAP estimators are generally based on the Rayleigh distribution [10], Gamma distribution [11], heavy-tailed Rayleigh distribution [26], and so on. Among these MAP estimators, the MAP estimator based on the Rayleigh distribution has a concise closed-form solution and, hence, has been the most commonly used.

The above despeckle algorithms aim at noise suppression and edge preservation but rarely take the integrity and saliency of targets into account. A detection-oriented despeckle algorithm is supposed to comprehensively solve all these problems.

C. Aircraft Detection

Traditional object detection algorithms for SAR images generally rely on statistical or geometric characterization. These algorithms possess excellent physical interpretability and pay great attention to the image characteristics. The constant false alarm rate (CFAR) detection algorithm [27], [28] is a classical object detection algorithm based on statistical characterization. The CFAR detection algorithm relies on statistical distributions to distinguish the targets from the background. It performs well in oceanic ship detection [29], [30] but is rarely applied for scenarios with cluttered and complex backgrounds, such as aircraft and building detection. Another type of traditional object detection algorithm relies on the geometric characterization [31], [32]. Chen et al. [31] designed a salient point vector (SPV) descriptor based on salient points of aircraft. The SPV descriptor achieves outstanding detection performance when the aircraft presents stable salient points on images. However, it is easily affected by the change in imaging conditions.

The deep learning detectors rely on the powerful feature extraction ability to achieve advanced performance and have become mainstream. The deep learning detectors can be divided into two-stage detectors and one-stage detectors. Two-stage object detection algorithms, such as Faster R-CNN [33], apply a region proposal network (RPN) to acquire the region of interest (ROI) and the corresponding probability of whether the ROI contains the target. The bounding box regression is performed to modify the coordinates of bounding boxes. Cascade R-CNN [34] extends the Faster R-CNN to a multistage detector. It selects samples with corresponding quality for different intersection-over-union (IoU) thresholds during the multistage training and detection. Cascade R-CNN detector, thus, achieves better recognition and localization. The one-stage detector such as YOLOX [35], [36] uses the deep semantic information and features of images to directly predict bounding boxes and categories without proposal generation. Compared with two-stage detectors, the one-stage detectors are easier to integrate and achieve better performance on processing efficiency due to the straightforward structure. The above deep learning detectors are based on predefined anchors or grids. There are also anchor-free algorithms that achieve outstanding performance. Center-Net [37] takes the object as a point and replaces the traditional bounding box label with a heatmap. It achieves outstanding performance both on 2-D and 3-D object detection. Sparse-RCNN [12] applies learnable proposal boxes instead of dense anchors to roughly search the objects and uses learnable proposal features to briefly describe the objects. Sparse R-CNN simplifies the detector with dense anchors and eliminates the process of nonmaximum suppression (NMS) at the same time. These deep learning models perform well on optical images but do not consider the special properties of SAR images. Their performance on SAR images is, hence, limited.

The aircraft detection algorithms based on deep learning methods for SAR images generally pay great attention to adapt the characteristics of SAR images with meticulously designed modules. The scattering feature relation network (SFR-Net) [38] is a typical deep learning detector that takes advantage of the salient points to locate and recognize the aircraft. It achieves superior detection performance by constructing geometric relationships among salient points of the aircraft with the attention mechanism. Guo et al. [39] acquired the

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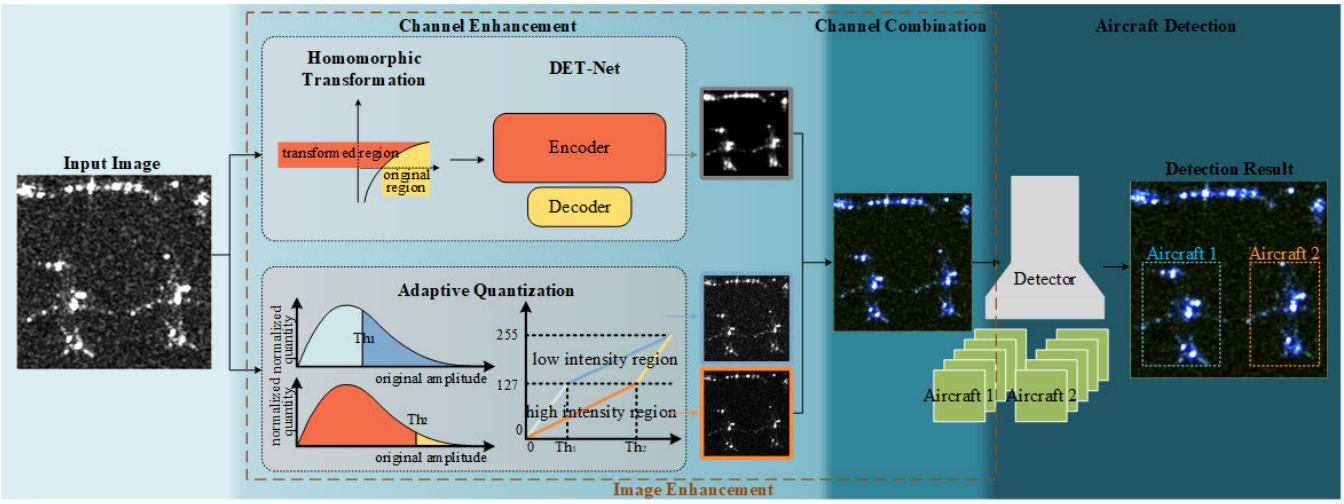


Fig. 1. Procedure of the aircraft detection based on the enhancement algorithm. The input image is processed with DET-Net and the adaptive quantization method based on unimodal distribution. The processed result is used for aircraft detection.

salient point features through a novel point detector and density-based clustering algorithm. The salient point features are processed with a neural network based on the multiscale feature extraction and attention mechanism. These aircraft detection algorithms achieve superior performance with the powerful feature extraction capabilities of neural networks. They make an effort to design novel detectors according to certain unique characteristics of SAR images but rarely pay much attention to the image quality problems that restrict aircraft detection. In addition, traditional image quality improvement methods usually only focus on a certain image quality problem and lack comprehensive optimization for aircraft detection. The mentioned issue limits the improvement of aircraft detection performance in SAR images.

III. Methodology

This section proposes an adaptive image enhancement algorithm for aircraft detection on SAR images. It collaboratively solves the drawbacks caused by the speckle effect and excessive dynamic range of SAR images. Both are almost the most common and serious problems impacting aircraft detection. The proposed image enhancement algorithm provides a pseudocolor image through channel optimization and combination for the aircraft detector. The pseudocolor image provides optimized visual features of aircraft through speckle suppression, such as a clearer outline, more complete shape, and more stable structure. The adaptive quantization method maintains more fine-grained radiation information and emphasizes effective visual features. By optimizing the input of the detector, the adaptive image enhancement algorithm further improves the detection performance.

The speckle noise suppression result is one of the enhanced channels obtained by DET-Net. DET-Net suppresses the speckle noise to improve the discrimination between the background and targets. The novel network also tries to maintain the structural features of targets, while most traditional despeckle algorithms rarely pay attention to this issue.

An adaptive quantization method based on the analysis of the amplitude distribution is used to maintain the fine-grained radiational features while compressing the dynamic range. The strong scattering feature enhancement channel and the weak scattering feature enhancement channel are the crucial components of the pseudocolor image, which enhances the radiational details of the targets and background, respectively.

The enhanced pseudocolor image is obtained by the combination of the enhanced channels. The detector finally employs the enhanced pseudocolor image to locate and recognize the aircraft. The detection procedure is shown in Fig. 1.

A. Mathematical Model

The RCS is a hypothetical surface that measures the intensity of scattered field [7], which can be described as follows:

$$\sigma = \left\| \sum_{k=1}^{n} \sqrt{\sigma_k} \cdot \exp\left(\frac{j4\pi d_k}{\lambda} \right) \right\|^2 \tag{1}$$

where n denotes the number of the isolated points contained in the minimum distinguishable unit, and σ and σ_k denote the module values of the RCSs of the minimum distinguishable unit and the isolated point, respectively. When the phase factor $(j4\pi d_k/\lambda)$ is randomly distributed in $[0, 2\pi)$, the RCS walks randomly in the complex space and the speckle noise problem arises accordingly. The module value of the RCS fluctuates severely due to the random walk. As a result, the speckle noise arises and destroys the textural and structural features of the aircraft.

This article employs the multiplicative model [7] to describe the observed amplitude as follows:

$$A = TN \tag{2}$$

where A denotes the observed amplitude, T describes the unspeckled amplitude, and N denotes the speckle noise. Different mathematical models are used to describe the amplitude distribution under different scenarios and conditions. Table I

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TABLE I
MATHEMATICAL MODELS OF AMPLITUDE DISTRIBUTION USED IN DIFFERENT SCENARIOS

Scenario	Mathematical Model
Farmlands	Nakagami Distribution [17]
Mountains	Fisher Distribution [20]
Forests	Lognormal Distribution [19]
Urban Areas(Low Resolution)	Inverse Gaussian Distribution [18]
Urban Areas(High Resolution)	Fisher Distribution

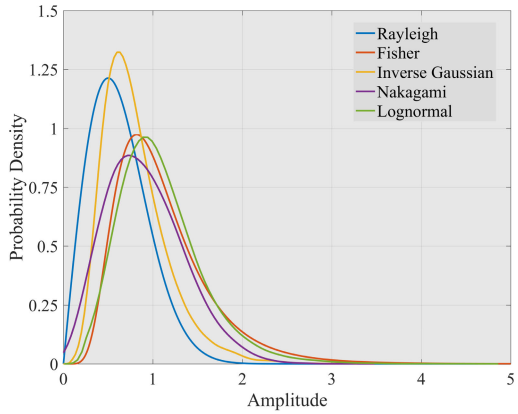


Fig. 2. PDFs of mathematical models commonly used.

shows the mathematical models of amplitude distribution commonly used in different observed scenarios, and Fig. 2 shows their pdfs. Although the specific mathematical models for the observed amplitude are different due to the observed scene and image resolution, they are all long-tailed unimodal distributions. According to the long-tailed unimodal distribution, it is possible to control the gray-scale distribution and emphasize the detailed information in certain amplitude intervals after quantization.

B. Despeckle Network

This part proposes a DET-net to suppress the speckle noise. The homomorphic transform is applied to the image to assist the despeckle network with convergence. Since there is no corresponding ground truth to guide DET-Net, a self-supervised learning method based on Bernoulli sampling [8] is applied for model training.

The speckle noise results in more severe amplitude fluctuation of the background and targets, which is reflected in the larger variance of the background and target areas. The severe amplitude fluctuation makes the background cluttered and interferes with the contour and edge of the aircraft. This problem reduces the discrimination between the background and targets and weakens the uniformity and integrity of visual features. The large amplitude variance of the aircraft causes the structural features of the aircraft more complex and random. Compared to the target with severe amplitude fluctuation, the target with low structural complexity and randomness is easier to capture and recognize by the detector.

DET-Net is designed as a codec network. The encoder aims to mine the latent representations of the input image and dilute the impact of the noise. The input image is divided into 256 patches and these patches are embedded into a 1024-D linear space. As shown in Fig. 3, the patch merging module employs the inverse pixel shuffle operation to deepen and downsample the feature maps. The inverse pixel shuffle operation rearranges the units in the neighborhood along the channel direction, which is shown in Fig. 3. The patch merging module enlarges the receptive field of each pixel in the feature map to enhance the holistic analysis of essential features. Furthermore, since the received signals are generally oversampled along both range and azimuth [41], the granularity of speckle noise is slightly larger than that of pixel units. The patch merging module splits the adjacent pixels, which makes the speckle noise much closer to the independent noise at the pixel level. After patch merging, the feature maps are further processed by the Swin Transformer module [42], which is shown in Fig. 3. The Swin Transformer module encodes the feature maps into latent representations through window attention and shift window attention mechanisms [42]. The decoder uses convolution operation to aggregate the latent representations and convert them into the image domain. The pixel shuffle operation finally upsamples the image to the original resolution, which is essentially the inverse operation of the patch merging.

A tricky problem for model training is lacking the guidance of ground truth since the definitive texture and structure of the aircraft are unavailable. To solve this problem, a self-supervised learning method based on the Bernoulli sampling [8] is applied for model training and inference. The image after the Bernoulli sampling with probability p is defined as follows:

$$\tilde{A}(x, y, q) = \begin{cases} A(x, y, q), & q \le p \\ 0, & q \ge p \end{cases} \tag{3}$$

where (x, y) denotes the pixel coordinate and q is a random number generated for each pixel based on the uniform distribution $U(0, 1)$. The essential task is to induce DET-Net to recover the original image based on the sampled image, which can be described by the following equation:

$$\begin{aligned} \Phi &= \operatorname{argmin}_{\Phi} E \| \left(\Phi\left(\tilde{A}\right) - A\right) * B \|_2 \\ B(x, y, q) &= \begin{cases} 1, & \tilde{A}(x, y, q) = 0 \\ 0, & \tilde{A}(x, y, q) \neq 0 \end{cases} \end{aligned} \tag{4}$$

where Φ is the ideal solution for DET-Net and $*$ denotes the Hadamard product. Quan et al. [8] employed (4) to reduce the variance of the image distribution and proved that such a self-supervised learning method is equivalent to guide the model with the image without noise when the noise is independent and additive [8]. The inverse pixel shuffle operation and Bernoulli sampling can ensure the speckle noise pixel independent. Since the self-supervised learning method based on Bernoulli sampling is not applicable to multiplicative noise, the image is processed with the homomorphic transform so that the corresponding noise is transformed into additive noise. The homomorphic transformation applies a logarithmic function, which is shown as follows:

$$H(A) = H(TN) = ln(TN) = ln(T) + ln(N) \tag{5}$$

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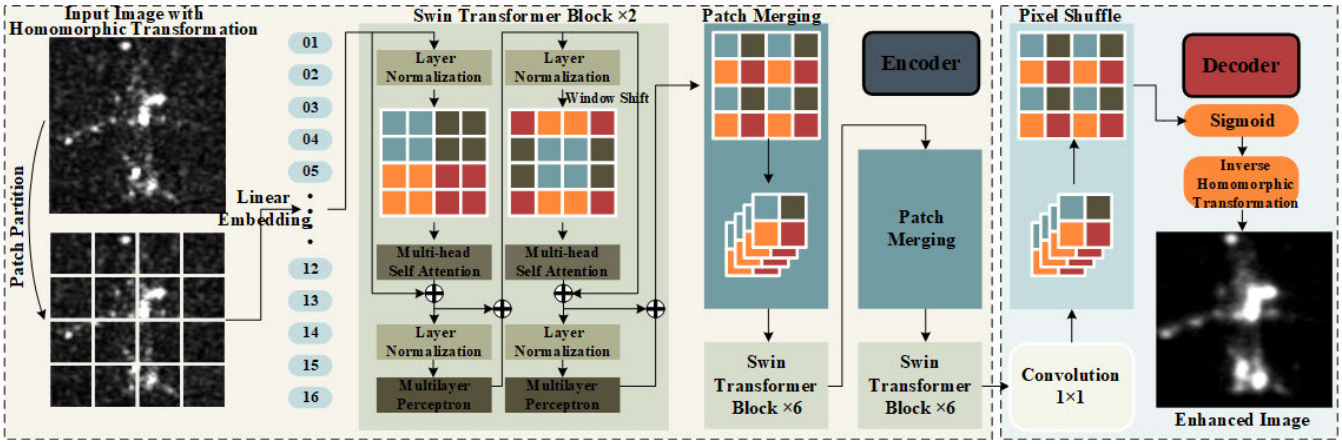


Fig. 3. Structure of the DET-net.

where $H(\cdot)$ denotes the homomorphic transformation. According to the Swerling model [43], [44], the RCS fluctuates with the incident angle. Due to the complex structure, the RCS of aircraft fluctuates severely and some components are even erased at certain incident angles. Since the erased components lose the ability to direct the network to reconstruct targets, they should not be involved in target reconstruction and loss calculation. An inpainting mask M based on a low threshold is used to mask the components of targets that are severely affected by the RCS fluctuation. As a result, the masked components are neither involved in image reconstruction nor in loss calculation. The mask threshold is calculated according to the unimodal long-tailed distribution and will be described in detail in Section III-C. The final loss function is given as follows:

$$\text{loss} = E\|\left(\Phi\left(\tilde{A} * M\right) - A\right) * B * M\|_2. \tag{6}$$

During inference, the image after Bernoulli sampling is fed to the trained codec network ten times, and the final enhanced result is the average of all the processed images.

C. Adaptive Quantization Method

The quantization operation is performed on the SAR image to emphasize effective information in the excessive dynamic range and reduce the storage burden before the detection procedure. The linear quantization method is widely used to compress the dynamic range. It emphasizes the effective features of amplitude truncation. However, it cannot maintain the stability of the gray-scale distribution, so the brightness and contrast of different images fluctuate severely. The logarithmic quantization method is another fundamental method. It converts the speckle noise into a simpler form but seriously decreases the radiational information in the high-scattering region. In this part, an adaptive quantization method is proposed to maintain the stability of the gray-scale distribution and preserve detailed information. The quantization procedure is shown in Fig. 4.

Taking the amplitude value P with the largest proportion in the unimodal amplitude distribution as a reference, the piecewise amplitude-gray mapping can be implemented to maintain the stability of the gray-scale distribution. Taking αP as the boundary, the amplitude values less than and greater than αP are, respectively, mapped to fixed gray-scale intervals to maintain the consistency of the converted gray-scale distribution. In this way, the brightness and contrast of different images are relatively stable after quantization, which is conducive to extract crucial features of the image. Setting different boundaries can emphasize the detailed information in

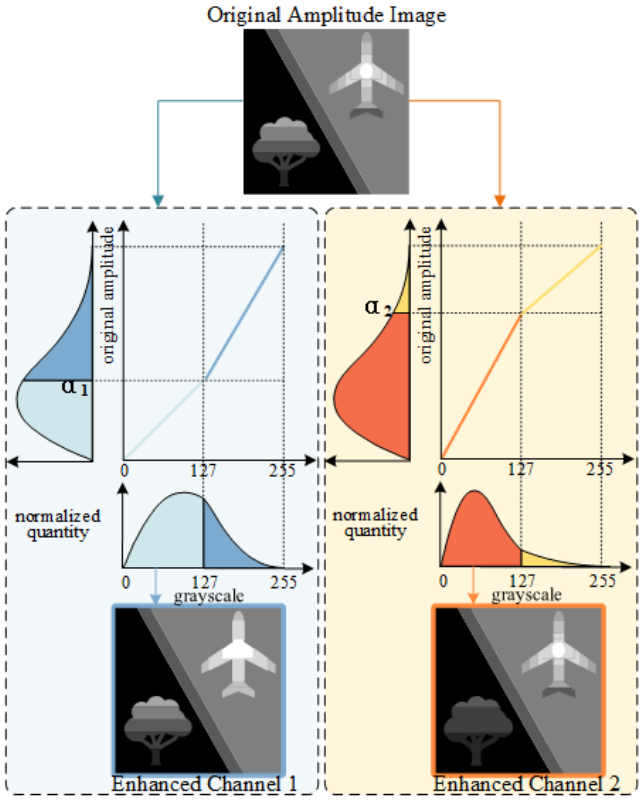


Fig. 4. Adaptive quantization method based on unimodal distribution. The weak and strong amplitudes are divided according to the threshold αP and mapped into the low and high gray-scale intervals, respectively. The left enhanced channel maintains the detailed information in part with low-amplitude values by applying a low threshold, while the right enhanced channel maintains the detailed information about the aircraft by applying a high threshold.

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different amplitude intervals. When a low boundary value is employed, the gray-scale range occupied by the background with lower amplitude values such as apron and vegetation is compressed, while targets with higher amplitude values such as aircraft are able to take advantage of a larger gray-scale range. On the contrary, more details of the background will be preserved. As shown in Fig. 4, the amplitude values below the boundary αP are mapped to the gray-scale interval of $[0, 127]$, while the other amplitude values are mapped to the gray-scale interval of $[128, 255]$. Both high and low boundary values are applied to the amplitude-gray-scale conversion, and the processed channels, respectively, preserve the detailed information in the parts with high- and low-amplitude values. The threshold of the inpainting mask M is also determined according to the value P. The threshold is set to ϵP, and ϵ denotes a very low percentage.

Although the scale and shape parameters of the amplitude distribution change with images, such a principle maps the amplitude intervals of the targets and background to relatively stable gray-scale intervals, so the gray-scale distribution of the processed image can stay consistent. The consistent brightness and contrast make the texture and structural features more stable, which helps the detector quickly converge to a superior solution. Different boundaries, respectively, maintain the detailed information of the target and the background. More detailed information means richer semantic features, which is conducive to the mining of semantic information and the improvement of detection performance.

DET-Net suppresses the speckle noise and provides a multiplicative noise suppression channel, which improves the discrimination between background and targets. The adaptive quantization method provides the strong scattering feature enhancement channel and the weak scattering feature enhancement channel, which improves the stability of visual features and emphasizes the details of the targets and background, respectively. The channel combination aggregates these channels into a pseudocolor image. The adaptive enhancement algorithm, thus, optimizes the input of the detector and improves the detection performance.

IV. Experiments and Analysis

The proposed algorithm applies channel enhancement and combination for the optimization of visual features and radiational details. Its ultimate goal is to improve the performance of detection and recognition of SAR images. Experiments are carried out to verify the despeckle ability of DET-Net and evaluate the adaptive quantization method. Ablation experiments are also provided to analyze the improvement in aircraft detection. The relative datasets and indicators are also introduced.

A. Dataset, Settings, and Indicators

Two datasets acquired by the GaoFen-3 satellite are applied for the experiments and analysis. One of them contains 25 single-polarization SAR images of three civil airports. It contains one category of aircraft, and the numbers of object instances in the trainset and testset are 4579 and 1315, respectively. The other dataset contains 41 single-polarization SAR images of 12 civil airports. It contains 12 types of aircraft, and the numbers of object instances in the trainset and testset are shown in Table II. Each dataset separates six images for evaluation. The trainsets and testsets with a size of 512×512 are obtained after sliding window processing. For the dataset with one category, the trainset covers 3258 aircraft slices and the testset covers 211 aircraft slices. For the other dataset with 12 categories, the trainset covers 3658 aircraft slices and the testset covers 610 aircraft slices. During the model optimization, 10% of the aircraft slices are randomly selected from each trainset as the validation set, and each detection algorithm takes the model with the best performance on the validation set as the test model. In addition, Faster RCNN uses ResNet-50 as the backbone. The Cascade RCNN algorithm consists of four stages, and the IoU thresholds of three detection stages are set to 0.5, 0.6, and 0.7. For Faster RCNN and Cascade RCNN algorithms, the initial learning rate is set to 0.005. YOLOX applies the middle-size model with the learning rate set to 0.025. Sparse RCNN applies ten learnable proposal boxes, and the initial learning rate is set to 10. DETNet is trained from scratch, the learning rate is fixed at 0.0001, and the training process lasts 500 epochs in total. DETNet and all the detection models are trained with the trainsets on a single NVIDIA RTX 2080 Ti.

TABLE II
Number of Object Instances

Category	Instance Number (Trainset)	Instance Number (Testset)
A220	1662	244
A320	824	121
A330	630	95
Boeing737	1088	150
Boeing747	226	38
Boeing767	914	160
Boeing777	180	40
G650	428	63
Airfreighter	226	33
Fokker-50	240	47
Helicopter	196	31
other	2308	338
total	8922	1360

The detection performance is evaluated on the recall, precision, F_1 score, and average precision (AP). The F_1 score and AP are defined as follows:

$$F_1 = \frac{2 * \text{Recall} * \text{Precision}}{\text{Recall} + \text{Precision}} \tag{7}$$

$$\text{AP} = \int_0^1 P(t)dt \tag{8}$$

where $P(t)$ denotes the precision varying with the confidence threshold. In addition, the IoU threshold is set at 50%, 60%, and 70% to calculate AP50, AP60, and AP70, respectively. The recall, precision, and F_1 score are all calculated at confidence threshold 50% and IoU threshold 50%.

B. Analysis of DET-Net

DET-Net is trained based on the Bernoulli sampling processing, and the sampling probability p directly affects the

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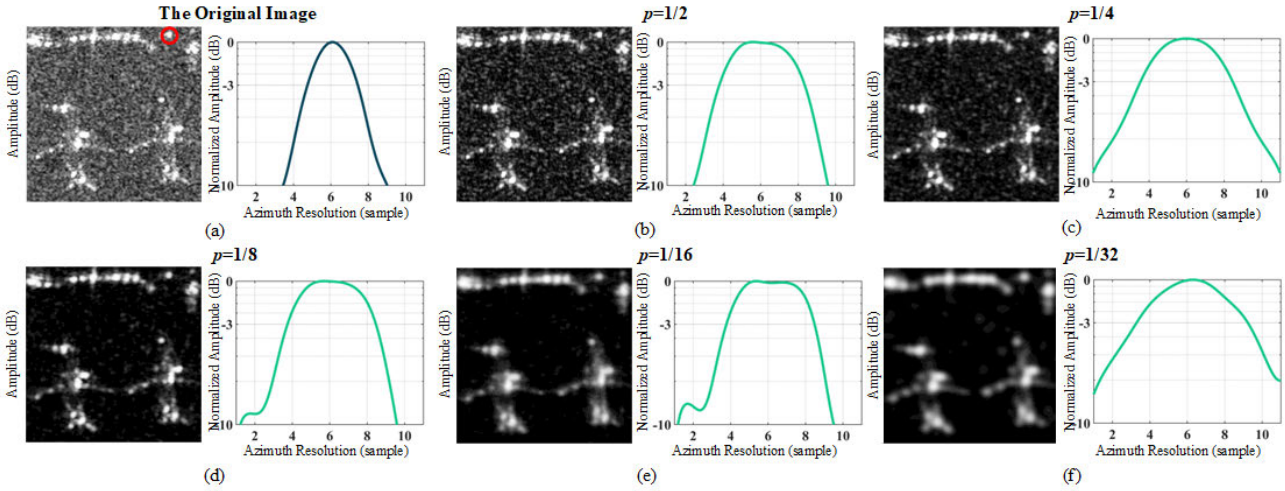


Fig. 5. Structure enhanced results with different sampling probabilities. (a) Displays the original image along with the corresponding azimuthal cross-section of the marked point. (b)–(f) Enhanced images with sampling probabilities of 1/2, 1/4, 1/8, 1/16, and 1/32, respectively, along with the corresponding azimuthal cross-sections of the same marked point.

TABLE III
VARIANCES WITH DIFFERENT SAMPLING PROBABILITIES

Sampling Probability	The Original Image	$p = 1/2$	$p = 1/4$	$p = 1/8$	$p = 1/16$	$p = 1/32$
Variance	665.198	146.248	124.182	115.358	61.242	141.388
Resolution	1.000 m	1.482 m	1.519 m	1.593 m	1.630 m	2.185 m

structure and texture-enhanced results. This part carries out experiments to define the optimal sampling probability p and evaluates the despeckle performance of DET-Net. There are many other speckle reduction algorithms that perform well in speckle noise suppression. These algorithms are also implemented as comparisons.

DET-Net mines the latent representations and rebuilds the image based on these latent representations. As the sampling probability becomes lower, the number of pixels that the reconstruction relies on decreases. In this case, the latent representations will cover more abundant and deeper semantic information to restore the image, but the restored image will lose more details. The multiplicative noise can be suppressed significantly, and the structure of the aircraft is also restored more completely. However, the image resolution also declines, which impacts the detection efficiency. Furthermore, if the sampling probability becomes extremely low, it will be impossible for DET-Net to restore the image. An extreme scenario is that when the sampling probability approaches 0, the network will lose the basis for reconstruction. An experiment is carried out to analyze the enhancement efficiency and resolution when different Bernoulli sampling probabilities are applied. As shown in (9), the variance of the homomorphic representation of the aircraft increases due to the noise, and the increment is proportional to the average power of the noise. The variance of the homomorphic representation of the aircraft is, hence, provided to evaluate enhancement efficiency

$$\begin{aligned} D[H(A)] &= D[H(N) + H(T_0)] \\ &= E[H(N) + H(T_0) - (E[H(N)] + E[H(T_0)])]^2 \\ &= E[H(N)]^2 + E[H(T_0)]^2 + 2E[H(N)H(T_0)] \\ &\quad - E^2[H(N)] - E^2[H(T_0)] - 2E[H(N)]E[H(T_0)] \\ &= E[H(N)]^2 - E^2[H(N)] + E[H(T_0)]^2 - E^2[H(T_0)] \\ &= D[H(N)] + \sigma^2_{H(T_0)}. \end{aligned} \tag{9}$$

Fig. 5 presents the enhanced results with different sampling probabilities, and each patch is under logarithmic quantization. The image resolutions along the azimuth and the variances of the homomorphic representation of the aircraft are provided in Table III. The image resolution is calculated according to the isolated point, which is marked by the red cycle in Fig. 5(a). The relative resolutions between the processed images and the original image are calculated by intercepting the 3-dB width of the shape of the point spread function [45], [46], [47]. Since the original image resolution is 1 m, the resolutions of the processed images can be calculated accordingly. Fig. 5 provides the point section of each processed image. Each variance shown in Table III is the average of 100 different aircraft. With artificially drawn masks of aircraft, the variances are calculated without the background.

The original image is seriously interfered with the speckle noise, and the aircraft structures are fragmented. Most parts of the aircraft are submerged in the speckle noise and cannot be distinguished. With the sampling probability decreasing, the speckle noise is suppressed more significantly, and the structure of the aircraft becomes complete and more stable. When the Bernoulli sampling probability is 1/16, the variance of aircraft reaches the minimum, and the speckle noise is almost eliminated. Although the image resolution declines, the top row of the point targets is still separated, and the resolution loss is relatively acceptable. When the Bernoulli sampling probability reaches 1/32, the variance of the aircraft increases greatly, and some parts of the aircraft become fuzzy and weak. The image resolution decreases catastrophically, and the top

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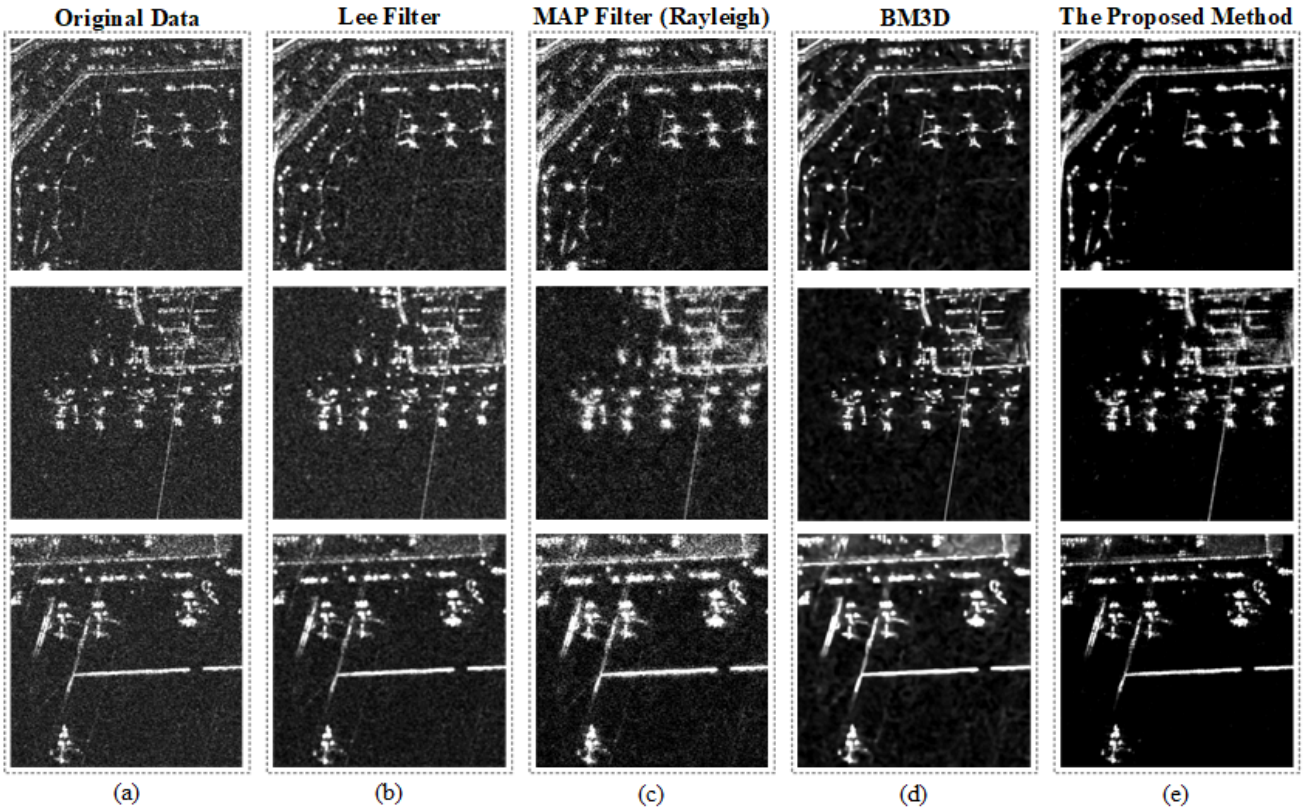


Fig. 6. Images processed with different despeckle algorithms. (a)–(e) Original images, Lee filter algorithm, Rayleigh MAP estimator, BM3D algorithm, and the proposed algorithm, respectively. All the images are presented under the logarithmic quantization.

TABLE IV
RESOLUTION AND PSNR OF EACH ALGORITHM

Algorithm	Original Data (without despeckling)	Lee Filter	MAP Filter	BM3D	Proposed Algorithm
PSNR	60.50 dB	63.55 dB	64.06dB	68.33 dB	70.53 dB
Resolution	1.000 m	2.192 m	1.423 m	1.0769 m	1.630 m

row of point targets stick together. The degradations of image quality originate from that such extreme sampling probability has already exceeded the tolerance of the model. Based on the above analysis, the proposed model can effectively suppress the speckle noise and enhance the features of targets. With comprehensive consideration of image resolution and feature enhancement effect, the Bernoulli sampling probability is set at 1/16.

Some advanced speckle suppression algorithms are implemented as comparisons to further confirm the ability of the proposed algorithm. Fig. 6 displays the processed results, and these images are presented under the logarithmic quantization. Table IV displays the resolution and peak SNR (PSNR) of each algorithm. The noise power can be evaluated in a homogeneous area [25], and the PSNR can be further calculated. The proposed algorithm achieves better performance than all these algorithms on despeckling, while BM3D achieves the highest resolution. Furthermore, the proposed algorithm improves the integrity and consistence of the features of aircraft, while BM3D makes the structure of each aircraft more broken and different.

These despeckle algorithms are applied for the data preprocessing, and their influences on aircraft detection can be further confirmed. Faster R-CNN is applied for the baseline, and both datasets mentioned in Section III-A are utilized for the experiments. The original images without despeckling and images with different despeckle algorithms have undergone linear quantization. Tables V and VI present the detection performance on different datasets. In Tables V and VI, the green numbers denote the indicators below the baseline, and the red numbers denote the best indicators. According to Table V, nearly all these speckle noise suppression algorithms improve the detection performance. Although BM3D achieves better resolution, its improvement in detection performance is not the most outstanding. The recall and AP60 even decrease after being processed with the BM3D algorithm. As for the more complicated scenario, the APs of all the categories are improved by these despeckle algorithms, as shown in

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TABLE V
DETECTION PERFORMANCE INDICATORS ACHIEVED BY DIFFERENT DESPECKLE ALGORITHMS

Algorithm	Recall	Precision	F1	AP50	AP60	AP70
Original data (without despeckling)	95.1%	85.4%	0.900	87.5%	86.4%	73.1%
Lee filter	95.0%	85.7%	0.901	87.5%	86.5%	73.4%
MAP filter	97.2%	86.1%	0.913	87.7%	86.4%	74.4%
BM3D	94.8%	86.9%	0.907	87.9%	86.3%	73.7%
Proposed algorithm	98.6%	87.6%	0.930	88.2%	87.1%	74.7%

TABLE VI
AP50 OF EACH CATEGORY ACHIEVED BY DIFFERENT DESPECKLE ALGORITHMS

AP50 Algorithm / Category	Original data (without despeckling)	Lee filter	MAP filter	BM3D	Proposed algorithm
A220	87.7%	89.1%	87.8%	89.1%	89.3%
A320	89.4%	90.4%	89.5%	90.0%	90.1%
A330	89.9%	90.9%	89.9%	90.9%	90.9%
Boeing737	89.9%	90.1%	89.4%	89.3%	90.2%
Boeing747	89.9%	90.7%	89.9%	90.0%	90.9%
Boeing767	89.7%	89.9%	89.1%	90.1%	89.9%
Boeing777	81.8%	81.8%	81.8%	81.8%	81.8%
G650	79.8%	83.8%	85.8%	85.8%	89.8%
Airfreighter	79.4%	82.7%	82.9%	85.7%	85.6%
Fokker-50	70.1%	80.1%	80.1%	80.1%	90.9%
Helicopter	50.3%	50.3%	53.2%	52.1%	70.6%
other	80.5%	80.7%	81.0%	81.0%	81.1%
average	81.5%	83.4%	83.4%	83.8%	86.8%

Table VI. The detectors rely on the structural and textual features to distinguish different types of aircraft. As the significance of the visual features becomes more remarkable, the proposed despeckle algorithm leads to more obvious improvements and increases the mean AP by 5.3%. Especially for small airplanes that are more easily submerged in noise, the proposed despeckle algorithm achieves the most remarkable improvements.

C. Analysis of the Adaptive Quantization Method

The normal quantization method is generally based on linear mapping, which linearly maps the amplitude region to the gray-scale region of [0, 255]. However, even for images of the same scene, the scale and shape parameters of the amplitude distribution change obviously. The gray-scale distribution, image brightness, and contrast after linear mapping, thus, become inconsistent and unstable, which makes aircraft detection more difficult. Fig. 7(a1) displays two images of the same scene, and Fig. 7(b1) displays corresponding gray-scale statistical histograms. Although these two images correspond to the same scene, their gray-scale distributions, image brightness, and contrast are quite different. Fig. 7(a2) and (a3), respectively, displays the first and second channels of the images processed with the proposed quantization method. Fig. 7(b2) and (b3) displays the corresponding gray-scale statistical histograms. To further evaluate the consistence of the gray-scale distributions after the numerical quantization, the KL divergences between these two images are provided in

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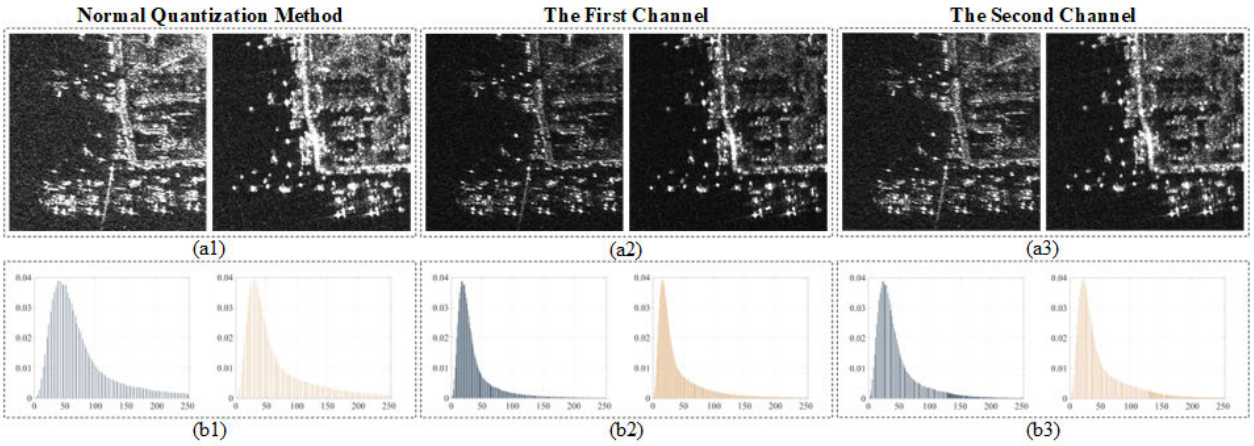


Fig. 7. Comparison of images and statistical histograms. (a1) Two images processed with normal quantization method. (b1) Corresponding gray-scale statistical histograms. (a2) and (a3) First and second channels of the images processed with the proposed algorithm. (b2) and (b3) Corresponding gray-scale statistical histograms.

TABLE VII
KL DIVERGENCE

Quantization Method	KL Divergence
Normal Quantization Method	3.8732
The Proposed Method (First Channel)	0.3017
The Proposed Method (Sirst Channel)	0.1506

Table VII. As shown in (10), the KL divergence is often used to measure the difference between two distributions $A_1(x)$ and $A_2(x)$. Generally speaking, the images with smaller KL divergence are more similar in gray-scale distribution, image brightness, and contrast

$$\mathrm{KL}(A_1||A_2) = \int A_1(x) lg \frac{A_1(x)}{A_2(x)} dx. \tag{10}$$

Compared with the images processed through the normal quantization method, the images processed with the proposed algorithm are more similar in gray-scale distribution, image brightness, and contrast. Experiments in Section III-D will verify that the proposed quantization method is able to significantly improve the detection performance.

D. Analysis of Detection Results

This part carries out ablation experiments to evaluate the detection performance. Through the analysis of the experimental results, the effectivity of the proposed channel enhancement and combination method can be confirmed.

The experiments based on the dataset with a single category are carried out to preliminarily analyze the performance of location and regression. Both anchor-based and anchor-free algorithms are applied for the verification. Besides the universal detection algorithms, SFR-Net, which is specially designed for aircraft detection in SAR images, is also implemented. The adaptive quantization method and the proposed despeckle algorithm are successively performed on the aircraft detection, and the corresponding detection effectivity is provided in Table VIII. The number of learnable proposal boxes of sparse R-CNN is set at 10, which is generally larger than the average number of aircraft in each pitch. Therefore, false alarms are maintained at a relatively high level, and the precision of sparse R-CNN is the lowest among these detectors.

According to Table VIII, the proposed adaptive quantization method and despeckle algorithm significantly improve the performance of all these detection methods. Among these employed detectors, Sparse R-CNN obtains the most obvious improvement on AP50 (2.7%) and Cascade R-CNN improves 6% on AP70 with the proposed algorithm. Since the proposed algorithm performs structural and radiational enhancement on the images and solves the problems that trouble aircraft detection, it can universally improve the performance of mainstream aircraft detection methods. The detection capacity is proved to be improved by the proposed algorithm, and it is necessary to further evaluate the performance under more complicated scenarios.

For a more comprehensive and convincing evaluation, ablation experiments based on the dataset with 12 categories of aircraft are implemented. The AP for each category is provided. The experiments are carried out on the Faster R-CNN algorithm. Both the linear and logarithmic quantization methods are implemented as the comparisons. The adaptive quantization method and the proposed despeckle algorithm are successively applied for the detection. The indicators shown in Table IX present the ability to capture and distinguish different types of aircraft.

The mean AP of the logarithmic quantization method is lower than that of the linear one by 0.9%. The logarithmic quantization method possesses a monotone-decreasing derivative of the intensity mapping. It, hence, mainly maintains the radiational details in the low scattering area and compresses the dynamic range of the strong scattering areas and targets. A more serious loss in the radiational details of aircraft leads to inferior detection performance. The channel combination method and the proposed despeckle algorithm give a general improvement in the AP of each category. As for the mean AP of all the categories, the channel combination method

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TABLE VIII
DETECTION PERFORMANCE

Algorithm	Recall	Precision	F1	AP50	AP60	AP70	Channel Combination	Despeckle with DET-Net
YOLOX	92.6%	86.6%	0.895	87.1%	85.6%	72.0%	✘	✘
	92.7%	87.0%	0.897	87.6%	85.5%	73.2%	✔	✘
	92.8%	88.2%	0.904	88.8%	85.8%	73.9%	✔	✔
Faster R-CNN	95.1%	85.4%	0.900	87.5%	86.4%	73.1%	✘	✘
	98.3%	85.3%	0.913	88.8%	87.8%	74.5%	✔	✘
	98.4%	87.9%	0.928	89.3%	88.7%	84.7%	✔	✔
Cascade R-CNN	96.7%	85.4%	0.913	87.8%	86.6%	74.0%	✘	✘
	97.6%	86.9%	0.919	88.8%	88.3%	84.9%	✔	✘
	99.3%	88.5%	0.936	90.1%	89.9%	88.0%	✔	✔
Sparse R-CNN	96.0%	26.2%	0.411	86.3%	84.2%	69.2%	✘	✘
	97.6%	26.6%	0.418	88.0%	86.5%	77.9%	✔	✘
	97.6%	26.6%	0.418	89.0%	88.2%	82.9%	✔	✔
SFR-Net	96.9%	87.2%	0.918	87.9%	86.4%	73.9%	✘	✘
	97.5%	87.6%	0.923	88.6%	88.0%	79.2%	✔	✘
	98.8%	88.9%	0.936	90.2%	89.0%	85.1%	✔	✔

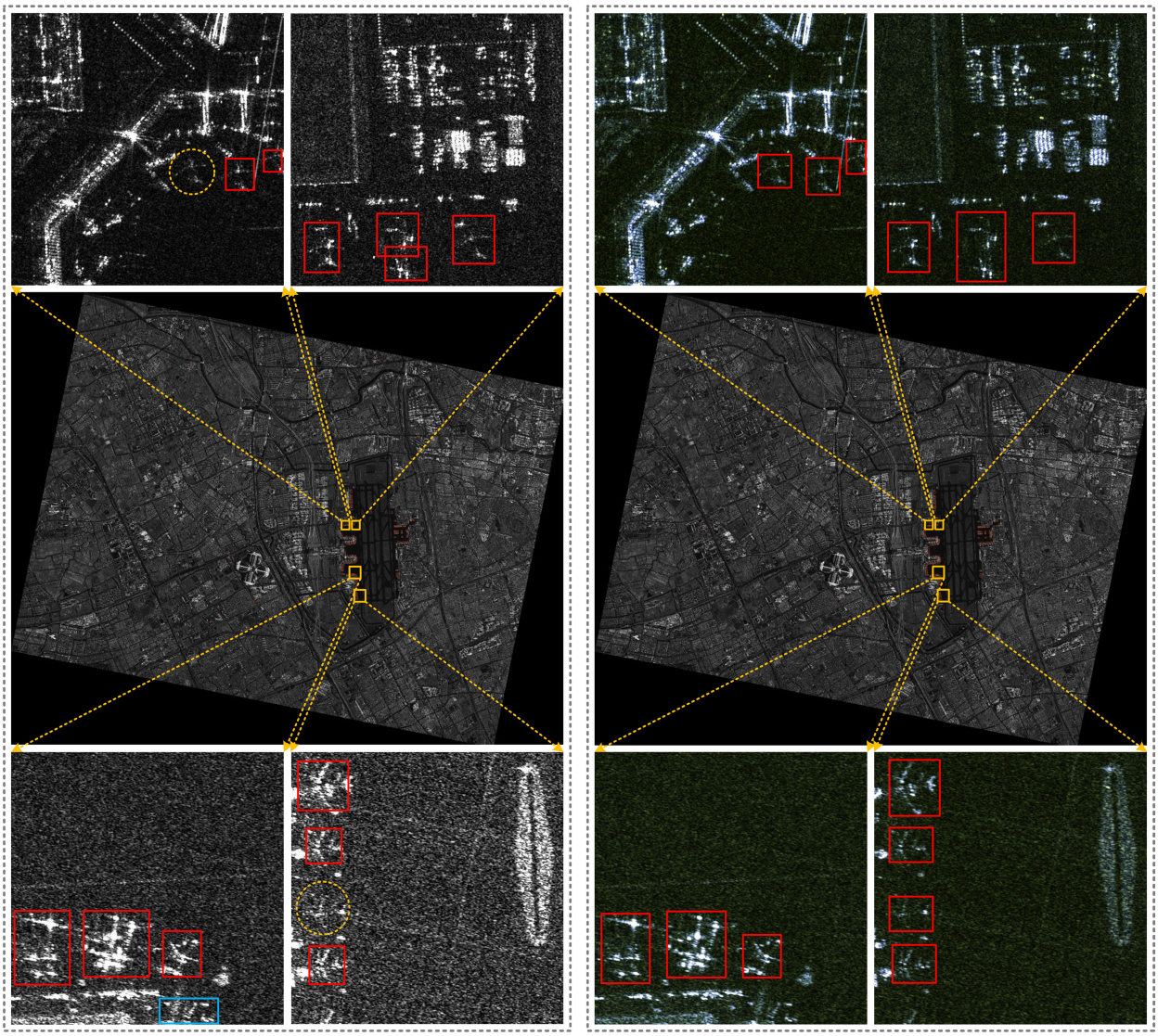
TABLE IX
RESULTS OF ABLATION EXPERIMENT (AP50)

AP50 Algorithm / Category	Linear Quantization	Logarithmic Quantization	Channel Combination	Combination & Despeckle
A220	87.7%	84.2%	89.3%	90.0%
A320	89.4%	89.2%	89.5%	90.3%
A330	89.9%	89.9%	90.9%	90.9%
Boeing737	89.9%	81.1%	89.9%	90.1%
Boeing747	89.9%	89.9%	90.9%	90.9%
Boeing767	89.7%	89.1%	90.6%	90.6%
Boeing777	81.8%	81.8%	90.3%	90.9%
G650	79.8%	80.1%	81.7%	90.8%
Airfreighter	79.4%	81.8%	82.1%	86.4%
Fokker-50	70.1%	70.1%	81.8%	90.4%
Helicopter	50.3%	50.3%	74.2%	81.8%
other	80.5%	79.5%	80.6%	81.4%
average	81.5%	80.6%	86.0%	88.7%

achieves an increase of 4.5%, and the proposed despeckle algorithm improves the mean AP by 2.7% on this basis. The above experiments indicate the proposed algorithm remarkably improves the detection capacity of the datasets. Next, a more comprehensive analysis based on the entire image is provided.

To verify the practical performance of the proposed algorithm, the Cascade R-CNN is applied to an entire view

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(a) (b)

Fig. 8. detection results on the entire view image. (a) Directly fed to the detector with sliding window detection process. The proposed algorithm is performed on each patch during the sliding window process in (b). Some representative patches obtained in the sliding window detection process are displayed, and the corresponding locations are marked in each image. In these patches, the dashed yellow circles and solid red rectangles, respectively, mark the aircraft miss and detected by the detector, and the blue rectangle marks the false alarm.

image in the test dataset. The patches obtained with the sliding process are actually applied for detection, and the detection results are painted on the original image.

Fig. 8(a) and (b) displays the detection results with the normal and proposed algorithms, respectively. The patches for detection are also displayed in Fig. 8. The second channel of the enhanced patches applies a lower division threshold than the first channel, so more pixels are mapped to the high gray-scale domain [128, 255] in the second channel. The overall brightness of the second channel is, hence, higher than that of the first channel. Since the first, second, and third channels, respectively, correspond to the red, green, and blue channels, most enhanced patches tend to be green on the whole. The third channel is filled with enhanced patches, on which the speckle noise is suppressed significantly. The proposed algorithm decreases the power of noise, so the target with a strong RCS is more prominent. Therefore, the targets such as the airplane in the images tend to be cyan and white, while the background such as the apron tends to be green. Such a difference in color makes the target more recognizable.

In Fig. 8, the dashed yellow circles and solid red rectangles, respectively, mark the aircraft miss and detected by the detector, and the blue rectangle marks the false alarm. The top left patches in Fig. 8(a) and (b) display the case where aircraft with extremely weak intensity are missed by the detectors. The bottom right patches in Fig. 8(a) and (b) show location interfered with extremely strong speckle noise, which seriously impacts the aircraft detection. In these cases, the targets are submerged in noise, and the contour and structural features are no longer obvious. Since the proposed algorithm suppresses the speckle noise and improves the SNR, the targets missed by the normal detection algorithm are captured. The false alarm

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enclosed with a blue solid rectangle has strong intensity and loses delicate gradation of gray-scale due to linear mapping, so it is easier to be mistaken with the aircraft. The proposed algorithm retains more details of this false target and decreases the intensity of some components, so it is distinguished from the aircraft. The aircraft in the top right patch in Fig. 8(a) is misjudged as two aircraft because of the broken structure. Since the third channel of the image processed with DET-Net improves the integrity of the target structure, the detector completely detects the entire target.

V. Conclusion

This article proposes a detection-oriented channel enhancement and combination algorithm for SAR images. The speckle noise and excessive dynamic range are the most common problems in SAR images, which seriously impact the extraction of visual features. The proposed algorithm improves detection performance by collaboratively solving these problems. DET-Net suppresses the speckle noise with a codec structure and provides clearer and more stable visual features. The adaptive quantization method emphasizes the effective information and preserves radiational details in the images. The channel combination method effectively integrates the enhanced channels into the pseudocolor image. The pseudocolor image provides a more consistent representation of visual features in similar scenes and strengthens the radiational details in regions of different scattering intensities. It, finally, assists detection algorithms in extracting more accurate localization and classification information. The experiments based on the datasets acquired by the GaoFen-3 satellite indicate that the proposed algorithm improves the performance of the detection and recognition. The novel enhancement algorithm achieves remarkable improvements on most of the mainstream detection algorithms, and the mean AP of all the categories increases by 7.2% after the proposed algorithm is applied for the detection.

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